

Peano Axioms

Lecture Notes on the Foundations of Natural Numbers

Mathematics Lecture Notes

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Abstract

These notes introduce the Peano axioms as a formal description of the natural numbers. The goal is not to make arithmetic more complicated, but to show what must be assumed in order to justify familiar operations such as addition, multiplication, order, and induction. The notes are written for students who already know elementary arithmetic, but who may not have seen mathematical foundations before.

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1 Why Do We Need Axioms for Natural Numbers?

At school we learn the natural numbers as a list:

$$0, 1, 2, 3, 4, \dots$$

This list is intuitive, but mathematics often requires more than intuition. We want to know which properties of these objects are fundamental and which ones can be proved from more basic assumptions.

Main idea

The Peano axioms describe natural numbers using only:

- a distinguished starting element, usually called 0,
- a successor operation S , which sends a number to the next number,
- the principle of mathematical induction.

The successor operation is the abstract version of “add one”. Instead of starting with the symbols $1, 2, 3, \dots$, we can define them by

$$1 = S(0), \quad 2 = S(1) = S(S(0)), \quad 3 = S(2),$$

and so on.

The Peano axioms answer questions such as:

- What does it mean that every natural number has a next number?
- Why is 0 not the successor of any natural number?
- Why can two different numbers not have the same successor?
- Why does induction prove statements about all natural numbers?
- How can addition and multiplication be defined rather than assumed?

2 The Basic Structure

2.1 The language of Peano arithmetic

At the most basic level, Peano arithmetic speaks about a structure

$$(\mathbb{N}, 0, S),$$

where:

- $\mathbb{N}$ is the set of natural numbers,
- $0 \in \mathbb{N}$ is the starting element,
- $S : \mathbb{N} \rightarrow \mathbb{N}$ is the successor function.

A more developed version of Peano arithmetic also includes addition and multiplication:

$$(\mathbb{N}, 0, S, +, \cdot).$$

However, it is conceptually useful to first understand 0 and S , because addition and multiplication can be defined recursively from them.

Definition 2.1 (Successor). The successor of a natural number n is denoted by $S(n)$. It is interpreted as the next natural number after n .

Informally:

$$S(0) = 1, \quad S(1) = 2, \quad S(2) = 3.$$

But formally, $1, 2, 3$ are only abbreviations:

$$1 := S(0), \quad 2 := S(S(0)), \quad 3 := S(S(S(0))).$$

3 The Peano Axioms

There are several equivalent ways to present the Peano axioms. In these notes we use the version based on 0 and the successor function.

Peano axioms

The natural numbers are governed by the following assumptions:

P1. 0 is a natural number.

P2. Every natural number has a successor: if $n \in \mathbb{N}$, then $S(n) \in \mathbb{N}$.

P3. 0 is not the successor of any natural number:

$$\forall n \in \mathbb{N}, \quad S(n) \neq 0.$$

P4. The successor function is injective:

$$\forall m, n \in \mathbb{N}, \quad S(m) = S(n) \Rightarrow m = n.$$

P5. Induction: if a set of natural numbers contains 0 and is closed under successor, then it contains every natural number.

The fifth axiom can be written in two common forms. The set-theoretic form is:

$$A \subseteq \mathbb{N}, \quad 0 \in A, \quad \forall n \in \mathbb{N} (n \in A \Rightarrow S(n) \in A) \quad \Rightarrow \quad A = \mathbb{N}.$$

The property form is:

$$P(0) \wedge \forall n \in \mathbb{N} (P(n) \Rightarrow P(S(n))) \quad \Rightarrow \quad \forall n \in \mathbb{N} P(n).$$

3.1 Meaning of each axiom

Axiom	What it prevents or guarantees
$0 \in \mathbb{N}$	There is a starting point.
Every n has $S(n)$	The counting process can always continue.
$S(n) \neq 0$	The chain does not loop back to zero.
$S(m) = S(n) \Rightarrow m = n$	Two different numbers cannot merge into the same next number.
Induction	There are no hidden natural numbers outside the chain generated from 0.

Important intuition

The first four axioms do not yet fully describe the usual natural numbers. They allow a structure that contains the usual chain

$$0, S(0), S(S(0)), \dots$$

and also some additional disconnected elements. The induction axiom eliminates these extra elements.

4 Induction as a Proof Method

The induction axiom becomes a practical proof technique.

Mathematical induction

To prove that a statement $P(n)$ is true for every $n \in \mathbb{N}$, prove two things:

1. Base case: prove $P(0)$.
2. Induction step: prove that for every n , if $P(n)$ is true, then $P(S(n))$ is true.

Then $P(n)$ is true for all natural numbers n .

4.1 A first example

Proposition 4.1. *For every $n \in \mathbb{N}$, if we define $1 = S(0)$, then*

$$n + 1 = S(n)$$

once addition has been defined in the usual recursive way.

This statement looks obvious, but in a formal development it follows from the recursive definition of addition. We first need to define addition, which we do in the next section.

4.2 Why induction works

Suppose that $P(0)$ is true and that truth passes from n to $S(n)$. Then:

$$P(0) \Rightarrow P(1) \Rightarrow P(2) \Rightarrow P(3) \Rightarrow \dots$$

Induction says that this chain reaches every natural number.

This last sentence is exactly where the Peano axiom is used. Without the induction axiom, there might be strange elements of $\mathbb{N}$ that are not reached by repeatedly applying S to 0.

5 Defining Addition from the Successor Function

Addition is usually treated as obvious. In Peano arithmetic, we can define it recursively.

Definition 5.1 (Addition). For fixed $n \in \mathbb{N}$, define $n + m$ by recursion on m :

$$n + 0 = n,$$

$$n + S(m) = S(n + m).$$

The first rule says: adding zero changes nothing. The second rule says: adding the successor of m means taking the successor after adding m .

Example 5.1. Let $2 = S(S(0))$ and $3 = S(S(S(0)))$. Then:

$$\begin{aligned} 2 + 3 &= 2 + S(2) \\ &= S(2 + 2) \\ &= S(2 + S(1)) \\ &= S(S(2 + 1)) \\ &= S(S(2 + S(0))) \\ &= S(S(S(2 + 0))) \\ &= S(S(S(2))) \\ &= 5. \end{aligned}$$

The calculation is longer than the school calculation, but it reveals that addition is not magic: it is repeated succession.

5.1 Basic theorems about addition

Lemma 5.1. *For every $n \in \mathbb{N}$,*

$$0 + n = n.$$

Proof. We prove the statement by induction on n .

Base case. For $n = 0$, we have

$$0 + 0 = 0$$

by the definition of addition.

Induction step. Assume that $0 + n = n$. Then

$$0 + S(n) = S(0 + n) = S(n),$$

where the first equality is the recursive definition of addition and the second equality uses the induction hypothesis.

Therefore $0 + n = n$ for all $n \in \mathbb{N}$. □

Lemma 5.2. *For all $n, m \in \mathbb{N}$,*

$$S(n) + m = S(n + m).$$

Proof. Fix n . We prove the statement by induction on m .

Base case.

$$S(n) + 0 = S(n)$$

by definition of addition, and

$$S(n + 0) = S(n),$$

so the equality holds for $m = 0$.

Induction step. Assume

$$S(n) + m = S(n + m).$$

Then

$$\begin{aligned} S(n) + S(m) &= S(S(n) + m) \\ &= S(S(n + m)) \\ &= S(n + S(m)). \end{aligned}$$

Thus the statement holds for $S(m)$. By induction, it holds for all $m \in \mathbb{N}$. □

Theorem 5.1 (Associativity of addition). *For all $a, b, c \in \mathbb{N}$,*

$$(a + b) + c = a + (b + c).$$

Proof. Fix a, b . We prove the statement by induction on c .

Base case.

$$(a + b) + 0 = a + b$$

and

$$a + (b + 0) = a + b.$$

So the statement is true for $c = 0$.

Induction step. Assume

$$(a + b) + c = a + (b + c).$$

Then

$$\begin{aligned}(a + b) + S(c) &= S((a + b) + c) \\ &= S(a + (b + c)) \\ &= a + S(b + c) \\ &= a + (b + S(c)).\end{aligned}$$

Therefore associativity follows by induction. $\square$

Theorem 5.2 (Commutativity of addition). *For all $a, b \in \mathbb{N}$,*

$$a + b = b + a.$$

Proof. Fix a . We prove the statement by induction on b .

Base case. We need $a + 0 = 0 + a$. The left side is a by definition, and the right side is a by the lemma $0 + n = n$. Hence $a + 0 = 0 + a$.

Induction step. Assume

$$a + b = b + a.$$

Then

$$\begin{aligned}a + S(b) &= S(a + b) \\ &= S(b + a) \\ &= S(b) + a,\end{aligned}$$

where the last equality uses the lemma $S(n) + m = S(n + m)$. Thus $a + S(b) = S(b) + a$. By induction, addition is commutative. $\square$

6 Defining Multiplication

Multiplication can also be defined recursively, using addition.

Definition 6.1 (Multiplication). For fixed $n \in \mathbb{N}$, define $n \cdot m$ by recursion on m :

$$n \cdot 0 = 0,$$

$$n \cdot S(m) = n \cdot m + n.$$

So multiplication is repeated addition, just as addition is repeated succession.

Example 6.1. Using the recursive definition,

$$\begin{aligned}3 \cdot 2 &= 3 \cdot S(1) \\ &= 3 \cdot 1 + 3 \\ &= (3 \cdot S(0)) + 3 \\ &= (3 \cdot 0 + 3) + 3 \\ &= (0 + 3) + 3 \\ &= 6.\end{aligned}$$

6.1 Basic properties of multiplication

From the recursive definition and induction, one can prove:

$$\begin{aligned} 0 \cdot n &= 0, \\ 1 \cdot n &= n, \\ n \cdot m &= m \cdot n, \\ (a \cdot b) \cdot c &= a \cdot (b \cdot c), \\ a \cdot (b + c) &= a \cdot b + a \cdot c. \end{aligned}$$

These are not primitive assumptions in this development. They are theorems.

Hierarchy of definitions

The Peano approach builds arithmetic in layers:

$$0, S \implies + \implies \cdot \implies \leq, <, \text{powers, divisibility, etc.}$$

7 Order on Natural Numbers

The order relation can be defined using addition.

Definition 7.1 (Order). For $a, b \in \mathbb{N}$, define

$$a \leq b$$

if and only if there exists $k \in \mathbb{N}$ such that

$$a + k = b.$$

We define

$$a < b$$

if and only if $a \leq b$ and $a \neq b$.

This definition captures the idea that b is obtained from a by moving forward a finite number of successor steps.

Example 7.1. We have $2 \leq 5$ because there exists $k = 3$ such that

$$2 + 3 = 5.$$

We do not have $5 \leq 2$, because there is no natural number k such that

$$5 + k = 2.$$

7.1 Basic order properties

The following facts can be proved from the Peano axioms:

- Reflexivity: $a \leq a$.
- Antisymmetry: if $a \leq b$ and $b \leq a$, then $a = b$.
- Transitivity: if $a \leq b$ and $b \leq c$, then $a \leq c$.
- Compatibility with addition: if $a \leq b$, then $a + c \leq b + c$.
- Compatibility with multiplication: if $a \leq b$, then $a \cdot c \leq b \cdot c$.

8 Strong Induction and the Well-Ordering Principle

The usual induction principle has an important equivalent form.

Theorem 8.1 (Strong induction). *Let $P(n)$ be a statement about natural numbers. Suppose that for every $n \in \mathbb{N}$,*

$$(P(0) \wedge P(1) \wedge \cdots \wedge P(n)) \Rightarrow P(S(n)).$$

Then $P(n)$ is true for all $n \in \mathbb{N}$.

More cleanly, strong induction is often written as:

$$\forall n \in \mathbb{N} \left([\forall k < n, P(k)] \Rightarrow P(n) \right) \Rightarrow \forall n \in \mathbb{N} P(n).$$

The intuitive meaning is: to prove $P(n)$, you may use all previous cases, not only the immediately previous one.

Theorem 8.2 (Well-ordering principle). *Every nonempty subset of $\mathbb{N}$ has a least element.*

Proof idea. Assume that a nonempty subset $A \subseteq \mathbb{N}$ has no least element. Then $0 \notin A$. If no number less than or equal to n belongs to A , and $S(n) \in A$, then $S(n)$ would be the least element of A . Therefore $S(n) \notin A$. By induction, no natural number belongs to A , contradicting that A is nonempty. $\square$

Equivalence of principles

For natural numbers, ordinary induction, strong induction, and the well-ordering principle are deeply connected. In many mathematical contexts, any one of them can be used to justify the others.

9 Recursive Definitions

The Peano axioms justify recursive definitions. This is essential because addition and multiplication were defined recursively.

Theorem 9.1 (Recursion principle, informal version). *Let A be a set, let $a \in A$, and let $f : A \rightarrow A$ be a function. Then there exists a unique function*

$$g : \mathbb{N} \rightarrow A$$

such that

$$g(0) = a$$

and

$$g(S(n)) = f(g(n))$$

for every $n \in \mathbb{N}$.

This theorem says that once we know where to start and how to pass from one step to the next, the whole sequence is determined.

Example 9.1 (A recursively defined sequence). Let $a_0 = 2$ and

$$a_{S(n)} = 3a_n + 1.$$

Then the sequence is uniquely determined:

$$a_0 = 2, \quad a_1 = 7, \quad a_2 = 22, \quad a_3 = 67, \dots$$

10 First-Order and Second-Order Peano Arithmetic

There is a subtle logical issue behind the induction axiom.

10.1 Second-order form

The set-theoretic induction axiom says:

$$\forall A \subseteq \mathbb{N}, \quad (0 \in A \wedge \forall n(n \in A \Rightarrow S(n) \in A)) \Rightarrow A = \mathbb{N}.$$

This quantifies over all subsets $A \subseteq \mathbb{N}$. For this reason, it belongs to second-order logic.

In second-order logic, the Peano axioms characterize the natural numbers uniquely up to isomorphism. Informally, every model is the same chain:

$$0, S(0), S(S(0)), \dots$$

10.2 First-order form

First-order Peano arithmetic cannot quantify over arbitrary subsets. Instead, it has an induction schema:

$$\varphi(0) \wedge \forall n(\varphi(n) \Rightarrow \varphi(S(n))) \Rightarrow \forall n \varphi(n),$$

for every formula $\varphi(n)$ expressible in the language.

This is not one axiom but infinitely many axioms, one for each formula φ .

A foundational warning

First-order Peano arithmetic has nonstandard models. These are models that satisfy all first-order Peano axioms but contain additional elements beyond the usual natural numbers. This does not make arithmetic useless. It shows that the logical language matters.

11 A Small Formal Development

This section summarizes how arithmetic can be built from the Peano axioms.

11.1 Step 1: names for small numbers

Define:

$$1 := S(0), \quad 2 := S(1), \quad 3 := S(2), \quad \dots$$

11.2 Step 2: addition

Define:

$$\begin{aligned} n + 0 &= n, \\ n + S(m) &= S(n + m). \end{aligned}$$

Then prove:

$$\begin{aligned} n + 0 &= n, & 0 + n &= n, \\ (a + b) + c &= a + (b + c), \\ a + b &= b + a. \end{aligned}$$

11.3 Step 3: multiplication

Define:

$$\begin{aligned} n \cdot 0 &= 0, \\ n \cdot S(m) &= n \cdot m + n. \end{aligned}$$

Then prove:

$$\begin{aligned} n \cdot 0 &= 0, & 0 \cdot n &= 0, \\ a \cdot b &= b \cdot a, \\ a \cdot (b + c) &= a \cdot b + a \cdot c. \end{aligned}$$

11.4 Step 4: order

Define:

$$a \leq b \iff \exists k \in \mathbb{N} (a + k = b).$$

Then prove the usual properties of order.

12 Common Misunderstandings

12.1 Misunderstanding 1: “The Peano axioms define numbers as symbols”

Not exactly. The axioms describe a structure. A model of the axioms consists of objects together with a distinguished object 0 and a successor function. The objects need not literally be written as 0, 1, 2, 3. What matters is the structure.

12.2 Misunderstanding 2: “Induction is just a proof trick”

Induction is not merely a trick. It is a structural principle saying that every natural number is obtained from 0 by finitely many applications of the successor function.

12.3 Misunderstanding 3: “Addition and multiplication are assumed”

In a foundational treatment, addition and multiplication can be defined recursively. Their familiar properties are then proved using induction.

12.4 Misunderstanding 4: “All versions of Peano arithmetic are the same”

There is a difference between first-order and second-order formulations. The second-order version is categorical under standard semantics. The first-order version is not categorical and has nonstandard models.

13 Examples for Lecture Discussion

Example 13.1 (Why injectivity matters). Suppose $S(a) = S(b)$. If the successor function were not injective, two different numbers could have the same next number. Counting would lose information: after moving one step forward, we would not know where we came from. Peano’s injectivity axiom prevents this.

Example 13.2 (Why zero must not be a successor). If $S(n) = 0$ for some n , the counting chain would loop back to the beginning. A simple example would be a clock-like structure:

$$0 \rightarrow 1 \rightarrow 2 \rightarrow 0.$$

Such a structure has successors, but it is not the natural numbers.

Example 13.3 (Why induction removes hidden elements). Imagine a structure that contains

$$0, 1, 2, 3, \dots$$

and also another chain

$$a, S(a), S(S(a)), \dots$$

not connected to 0. The first four axioms may still hold. But induction fails: the set generated from 0 contains 0 and is closed under successor, yet it does not contain a . Therefore this structure is not allowed by the Peano induction axiom.

14 Exercises

Exercise 14.1. Using the recursive definition of addition, compute $2 + 2$ purely in terms of successor.

Exercise 14.2. Prove by induction that $0 + n = n$ for all $n \in \mathbb{N}$.

Exercise 14.3. Prove that $n + 1 = S(n)$ for all $n \in \mathbb{N}$, where $1 = S(0)$.

Exercise 14.4. Using the recursive definition of multiplication, compute $2 \cdot 3$.

Exercise 14.5. Prove that $0 \cdot n = 0$ for all $n \in \mathbb{N}$.

Exercise 14.6. Define $a \leq b$ by $\exists k \in \mathbb{N} (a + k = b)$. Prove that $a \leq a$ for every $a \in \mathbb{N}$.

Exercise 14.7. Explain in words why the induction axiom is needed in addition to the successor axioms.

15 Short Summary

What students should remember

- The Peano axioms formalize the natural numbers using 0, successor, and induction.
- The successor function represents the operation “go to the next number”.
- The axioms prevent loops, merging successors, and hidden elements.
- Addition is defined as repeated succession.
- Multiplication is defined as repeated addition.
- Induction is not an optional trick: it is the core structural principle of natural numbers.
- First-order and second-order Peano arithmetic are logically different.

16 Suggested Board Plan for a 90-Minute Lecture

Time	Content
0–10 min	Motivation: why natural numbers need axioms.
10–25 min	The structure $(\mathbb{N}, 0, S)$, successor, definitions of 1, 2, 3.
25–40 min	Peano axioms and explanation of each axiom.
40–55 min	Induction as an axiom and as a proof method.
55–70 min	Recursive definition of addition; prove $0 + n = n$.
70–80 min	Multiplication and order as derived notions.
80–90 min	First-order versus second-order comments; summary and questions.

Final Perspective

The Peano axioms are a clean example of how mathematics builds familiar objects from minimal assumptions. Natural numbers are not introduced by drawing infinitely many symbols on the board. They are characterized by a starting point, a successor operation, and an induction principle. From these ingredients one can reconstruct arithmetic: addition, multiplication, order, recursive sequences, and the usual proof methods of elementary number theory.